



FAKULTI SAINS KOMPUTER DAN MATEMATIK (FSKM)

SEMESTER II 2025/2026

CSE3433 SOFTWARE ARCHITECTURE (K2)

Deliverable 2: Architecture Design

GROUP 8

Service-Oriented Smart Campus Support System

(Service-Oriented Architecture)

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1.0 System Overview

The Service-Oriented Smart Campus Support System is a centralized digital ecosystem that is designed to replace the divided and disorganised infrastructures that currently make it hard for universities to run smoothly. By using the Service-Oriented Architecture (SOA), the platform will combine all of the isolated functions such as academic registration, resource scheduling and administrative support into a single and interoperable hub.

This transition from isolated platforms to an integrated service model that can eliminate data redundancy, strengthens security through centralized identity management and provides a smooth user experience for both students and faculty.

1.1 Problem Description

Modern university campuses face challenges in managing student services due to fragmented and poorly integrated systems. Students and staff often use multiple platforms for tasks like course registration, facility booking, issue reporting, and receiving updates, which leads to inconvenience, duplicated data, and delays in accessing services.

For instance, course registration may lack real-time validation for prerequisites or seat availability, while facility bookings can cause scheduling conflicts without a centralized system. Issue reporting is also inefficient, with limited visibility on request status. Additionally, the absence of a unified identity and access control system creates security risks and inconsistent authorization across services. Weak communication between systems further results in delayed or missed updates.

To address these issues, a Service-Oriented Smart Campus Support System is proposed. By using Service-Oriented Architecture (SOA), the system integrates key services such as Identity Management, Academic Management, Facility Booking, Helpdesk, and Notification into a modular and interoperable structure. This approach improves efficiency, ensures data consistency, enhances user experience, and supports future scalability.

1.2 System Objectives

The purpose of this system is to:

- Provide a platform that allows students and staff to access campus services in one system without having to use separate systems.
- Improve the efficiency of campus facility management through a more organized system. Duplication or errors will be reduced.
- Facilitate communication between users and the system through an automatic notification and alert system. Users will get the latest information in real-time.
- Develop a flexible and easily expandable system using service-oriented architecture (SOA) approach that allows the system to be improved or expanded easily without affecting the entire system.

2.0 Functional Requirements

The Service-Oriented Smart Campus Support System is divided into five initial functions. Each function is designed to be autonomous while providing users the specific capabilities via RESTful API endpoints.

2.1 Identity & Access

This function acts as the important authority for user verification and role management in the system.

- FR-101: The system shall allow students and staff to log in using their unique Username and password.
- FR-102: The service shall provide an endpoint for other services to verify if a user has "Student" or "Admin" privileges.
- FR-103: The system shall provide an endpoint to validate if a user's session is still active before allowing access to other services.
- FR-104: The system shall retrieve user profile data (Name, Student ID, and Role) upon request from authorized services.

2.2 Academic & Course

This service manages the academic catalog and the enrollment process for students.

- FR-201: The system shall provide a list of available courses, including course codes, names, and credit hours.
- FR-202: The system shall allow students to register for specific courses during the open registration period.
- FR-203: The service shall verify if a student meets the requirements for a course before allowing registration.
- FR-204: The system shall automatically close registration for a course once the maximum number of students is reached.
- FR-205: The service shall verify the student's active status via the Identity function before finalizing any course enrollment.

2.3 Facility Booking

This service manages the digital reservation of physical campus resources like discussion rooms and computer labs.

- FR-301: The system shall display a list of available rooms (Discussion Room A, Lab 1) along with their availability status.
- FR-302: Students shall be able to create, view, and cancel reservations for specific time slots.
- FR-303: Conflict Prevention: The system shall validate that a room does not have another booking for the same time period.
- FR-304: The system shall communicate with the Identity Service to verify the student's eligibility before confirming a reservation.

2.4 Campus Support & Helpdesk

This function handles the technical and administrative requests and allows students to report issues digitally.

- FR-401: Users shall be able to submit an issues report with a title, description, and priority level (Low, Medium, High).

- FR-402: Students shall be able to track the progress of their issues report("Pending," "In Progress," "Resolved").
- FR-403: Staff shall be able to assign the issues report to specific technicians for resolution.

2.5 Notification

This utility function will handle the communication and user feedback.

- FR-501: The system shall trigger an automatic "Booking Successful" alert once a reservation is recorded in the facility booking functions.
- FR-502: The system shall notify the student whenever the status of their support ticket changes (from "Pending" to "Resolved").
- FR-503: The system shall provide a real-time notification to Administrators when a new facility is added or a critical issues report is submitted.
- FR-504: The system shall alert students when the registration deadline is approaching.

3.0 Non-functional Requirements

This section will outline the quality attributes and some of the operational constraints for the Service-Oriented smart Campus System.

3.1 Performance

- NFR-101 : The system shall support at least 5,000 concurrent users with an average response time less than 2 seconds during peak periods such as course registration weeks.

3.2 Security

- NFR-201 : The system shall implement Centralized Role-Based Access Control (RBAC) via the Identity Management service to restrict access to sensitive academic and administrative data.

3.3 Usability

- NFR-301 : The UI must be fully responsive, ensuring a seamless experience across all of the web browsers, tablets and mobile devices.

3.4 Reliability & Maintainability

- NFR-401 : The failure of one non-critical service such as Helpdesk shall not cause the failure of other services like Identity Management.
- NFR-402 : Automated daily backups of all databases must be maintained to ensure zero data loss in the event of a system failure.

3.5 Availability

- NFR-501 : The system shall maintain an uptime of 99.9% with 24/7 availability, excluding scheduled maintenance windows.

3.6 Scalability

- NFR-601 : The system shall be capable of scaling horizontally to accommodate a growth of up to 10,000 active users as the university expands.

4.0 Architecture Style Explanation

This Smart Campus Support System will be designed using the RESTful web services within a Service-Oriented Architecture to break down the systems into five autonomous and loosely coupled services such as identity, academic, booking, helpdesk and notification.

4.1 Key Design Principles

The system follows the Service-Oriented Architecture principles in the following things:

- **Statelessness:** Every time the service gets a request it has all the information it needs to process that request. This will help the system work well with a lot of users and makes it easier to manage the servers
- **Loose Coupling:** The services can work on their own or independently so that the user can update the Facility Booking part without messing with other parts like the

Identity or Academic parts because of the minimal dependency it has.

- **Standardized Contracts:** All the services communicate with each other in a way using something called REST API interfaces and a format called JSON. This helps all the different parts of the system work together smoothly.
- **Service Autonomy:** Each service controls its environment, which helps the whole campus system work better.

4.2 Service Classifications

To ensure the business alignments and technical efficiency, the services are categorized as follows:

- **Business Services:** These services represent the main and the core of the campus activity such as the academic management for course registration and facility booking for the room reservations.
- **Utility Services:** These services offer platform-wide technological capabilities. For example, Identity and Access need instantaneous replies, but Notification is asynchronous.
- **Entity Services:** This service will act as the bridge to the database for handling simple tasks for specific objects like student profiles or room details.

5.0 Service or Component Identification

This section explains how the Service-Oriented Smart Campus Support System is logically divided into separate, manageable and very integrated services. In alignment with Service-Oriented Architecture (SOA) principles, the system is broken down into separate business services which each have their own set of specific functional capabilities.

5.1 Core Service Domains

There are five main service domains in the architecture which have specialised parts in each domain that are in charge of certain business processes.

5.1.1. Identity and Access Management Service

This service is like an authority for user login and permission across the campus network to make sure that access is safe and that users only need to sign in once.

- Authentication Component: Checks user details against the university database.
- Authorization (RBAC) Component: Controls who can use different services by managing user roles such as the “Undergraduate”, “Faculty” or “Admin”.
- Profile Management Component: Handles the retrieval and updating of basic user demographic and contact information.

5.1.2. Academic Management Service

This service encapsulates all logic related to student academic progression by only focusing heavily on checking in time to avoid registration mistakes.

- Course Registration Component: Processes student enrollment and updates their academic records.
- Prerequisite Validation Component: Checks academic data to see if a student meets course criteria.
- Timetable Integration Component: Creates a schedule for enrolled courses that has conflict-free for the student.

5.1.3. Facility and Resource Service

This service manages the scheduling and allocation of physical campus assets in order to make sure the optimal utilization and prevents any booking conflicts.

- Availability Checker Component: Provides real-time status of physical spaces like the lecture halls, computer labs and the equipment.
- Booking Engine Component: Processes reservation requests, manages time slots and handles any booking modifications or cancellations.
- Conflict Resolution Component: Validates new booking requests against existing reservations to prevent double-booking.

5.1.4. Helpdesk and Support Service

A centralized routing system for all administrative, IT and maintenance requests.

- Ticket Generation Component: Allows users to log issues and categorise them by department or urgency.
- Status Tracking Component: Tracks the lifecycle of a support ticket and provides updates to the user.
- Routing Component: Automatically assigns tickets to the appropriate campus department based on the issue category.

5.1.5. Notification and Alert Service

A shared utility service utilized by all other domains to push asynchronous communication to users.

- Event Listener Component: Monitors other services for specific triggers such as a changed ticket status, a successful facility booking or a system-wide emergency.
- Message Dispatcher Component: Formats and routes messages through the appropriate channels such as in-app push notifications or email.

6.0 System Decomposition

Details the decomposition of the system into services that have a single responsibility in line with SOA principles.

Service Name	Responsibility	Exclusions
Identity & Access	Manage Single Sign-On (SSO), Role-Based Access Control (RBAC), session authentication, and user profile management.	Does not keep student academic records or room reservation logs.
Academic & Course	Manage course catalogs, registration logic, and subject prerequisite verification.	Does not handle user password resets or facility maintenance.
Facility & Resource	Manage real-time room availability, booking engine, and schedule conflict resolution.	Does not manage student Grade Point Average (GPA) or technical support tickets.
Helpdesk & Support	Manages complaint ticket creation, issue status tracking, and department routing.	Does not process course registration or user identity verification.
Notification	Acts as a utility to trigger notifications based on events in the system.	Does not own core business data, only processes message templates and dispatches.

7.0 Boundaries

The definition of boundaries for each component is critical to ensure loose coupling between services in the Smart Campus Support System:

- **Identity & Access Service:** The boundaries of this service are limited to verification of the identity and role of the user only. It **DOES NOT** have the authority to determine a student's academic eligibility for subject enrolment.
- **Academic Service:** Fully responsible for the integrity of academic and registration data. It is **NOT ALLOWED** to send notifications directly to users instead, it is necessary to go through the Notification Service for any communication.
- **Facility & Resource Service:** To manage the physical assets of the campus. It **DOES**

NOT store or process financial information or student fee debt.

- **Helpdesk & Support Service:** Focus on complaint management and technical support. This service **DOES NOT** have access or permission to alter any records of students grades or achievements in the academic database.
- **Notification Service:** Acts as a single agent. It **DOES NOT** have the authority to permanently store complaint history or academic details. The service also cannot determine who should receive the notification.

8.0 Communication Design

Protocol and interaction pattern used to ensure the system's modularity and performance.

8.1 Communication Protocols.

The system utilizes three distinct communication methods to balance real-time data needs with system reliability:

- **Synchronous RESTful APIs (REST):** Used for primary operations where an immediate response is required, such as login verification and course registration .
- **Asynchronous Event-Driven Communication (Events):** Facilitated by the Notification Service's "Event Listener" to handle background tasks like sending emails without slowing down the user interface.
- **Layered Database Calls:** Direct SQL-based communication between individual service modules and the MySQL database to ensure data persistence.

8.2 Interaction Mapping.

Based on the **SOA principles** of loose coupling and service autonomy, the interactions are mapped as follows:

Source	Target	Protocol	Purpose
Client Web Browser	API Gateway	HTTPS	Initial entry point for all user requests.
Academic Service	Identity Service	REST API	To verify student active status before enrollment.
Facility Service	Identity Service	REST API	To verify student eligibility for booking
Business Services	Notification Service	Event Trigger	To alert the listener that a process (e.g., booking) is complete.

8.3 Implementation Strategy

To maintain the Stateless and Loose Coupling mentioned in your architecture style explanation:

1. **Request Flow:** All external requests must be intercepted by the **API Gateway** for routing and security before reaching the core services.
2. **Data Isolation:** Each service communicates with its own logical database to prevent data redundancy and ensure systematic segregation.
3. **Real-time Interaction:** Use the **Interaction Diagrams** (Figures 6.3–6.6) as the logic flow for coding the `saveIssue()`, `fetchCourses()` and `validateUser()` methods in your server .

9.0 Data Ownership

The pattern “Database-per-Service” is used in Service-Oriented Architectures to make sure that each service is the single source of truth for its own domain. No service can ever touch a different service database. All communication must happen over the defined RESTful APIs or Event triggers.

The data ownership boundaries in the Service-Oriented Smart Campus Support System are defined in the following table:

Service Domain	Primary Data Entities Owned	Ownership Responsibilities and Constraints
Identity & Access	● User Credentials (Passwords/Hashes) ● User Profiles (Name, Contact Info) ● Roles (Student, Admin, Staff) ● Session Tokens	Serves as the final authority for user authentication. It does not retain academic grades, booking history or support tickets.
Academic & Course	● Course Catalogs ● Student Enrollment Records ● Academic Timetables ● Credit ● Hours	The parent record for all academic progression. It cannot have the user names or active status. It must call the Identity Service through the API to get that info.
Facility & Resource	● Physical Asset Inventories (Rooms, Labs, Equipment) ● Booking Schedules ● Availability Status ● Capacity Limit	Manages physical space lifecycle and prevents scheduling conflicts. It only stores the “Student ID” of the person who booked the room, and fetches the actual profile from the Identity Service.
Helpdesk & Support	● Support Tickets ● Issue Categories ● Priority Levels ● Technician Assignments ● Resolution Status Logs	Life cycle of administrative and technical complaints. It can not alter academic records or the condition of facilities.
Notification	● Message Templates ● Delivery Logs (Sent/Failed) ● Notification Triggers ● User Communication Preferences	Acts mainly on temporary data. It records what messages went to whom but it doesn’t own the core business data (such as the specifics of a booked room) that triggered the alert, permanently.

10.0 Architecture Diagrams

10.1 System Context Diagram

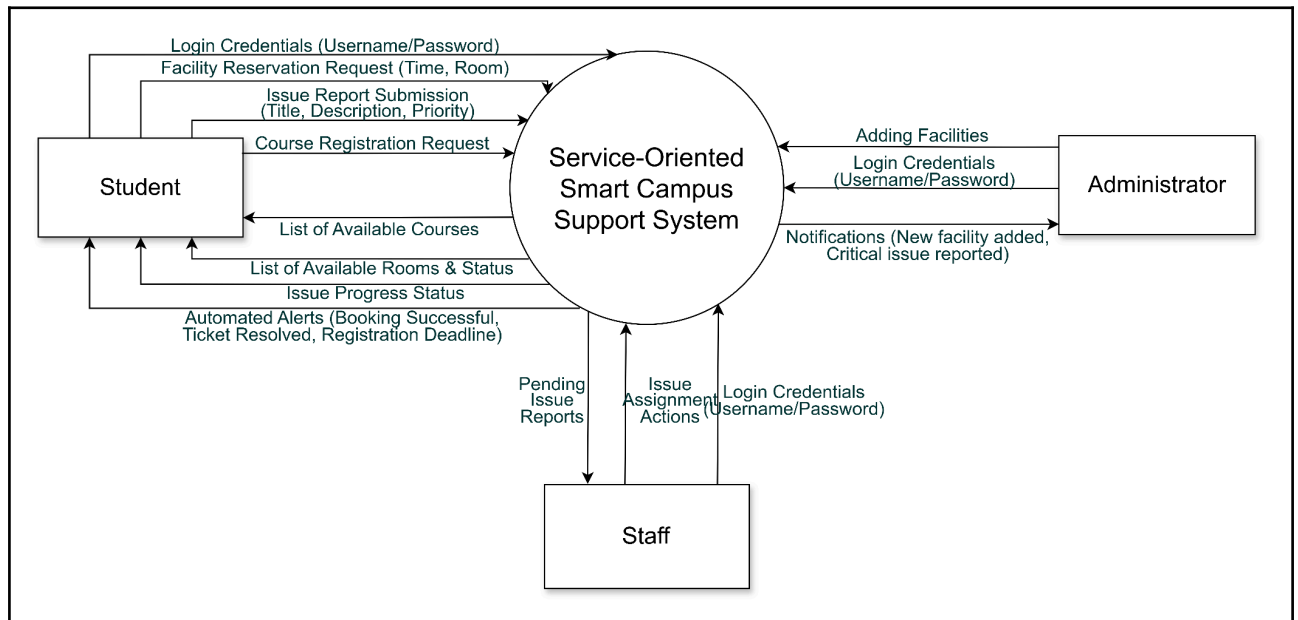


Figure 10.1: System Context Diagram for Service-Oriented Smart Campus Support System

The System Context Diagram in Figure 6.1 above shows the boundary of the Service-Oriented Smart Campus Support System with its context. The system interacts with three primary external entities that are from Students, Staff and Administrator. The internal workings of the system such as the databases and individual services will be hidden in the center which is called "black box" to focus on only what data should go in and what need to come out.

10.2 Component /Service Architecture Diagram

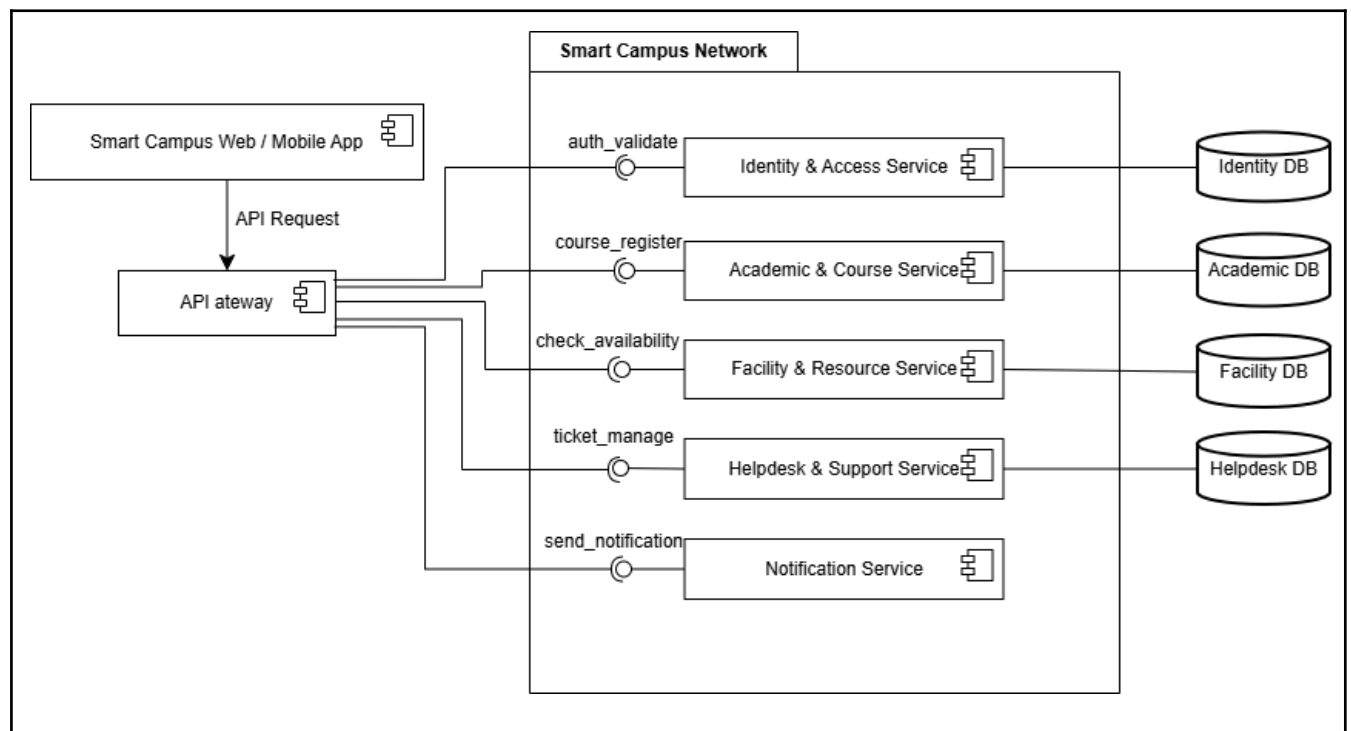


Figure 10.2: *Component /Service Architecture Diagram for Service-Oriented Smart Campus Support System*

The Component /Service Architecture Diagram in Figure 10.2 shows the logic of the Service-Oriented Smart Campus System based on the principles of loose coupling and service autonomy. API Gateway acts as a single mediator that receives requests from client applications and routes traffic to the relevant domain services through predefined standardized interfaces contracts. Each service, such as Identity & Access and Academic & Course, has its own database to ensure Data Ownership principles are followed, while reducing redundancy and improving data security in the campus network. Overall, this design allows each module to function independently while remaining seamlessly integrated through a secure intermediate layer.

10.3 Interaction Diagram

The interaction diagrams illustrate how different services within the *Service-Oriented Smart Campus Support System* work together to handle key user activities such as issue reporting, course registration, facility booking, and administrative management. In all interactions, the process begins with user authentication through the Identity & Access Service to ensure secure and role-based access control.

Once verified, users can perform specific actions depending on their role. Students are able to register for courses, book facilities, and submit issue reports, while administrators can manage resources such as adding new facilities. Each request is processed by its respective service, such as Academic, Facility Booking, or Helpdesk, where validations like prerequisite checking, conflict detection, and data verification are performed before storing the information in the database.

Throughout these processes, the Notification Service plays an important role by sending real-time updates, such as booking confirmations or issue status changes. This integrated interaction between services demonstrates the advantages of the Service-Oriented Architecture (SOA), including improved efficiency, reduced data redundancy, better system coordination, and a more seamless user experience compared to traditional fragmented systems.

10.3.1 Interaction Diagram for Issue Reporting

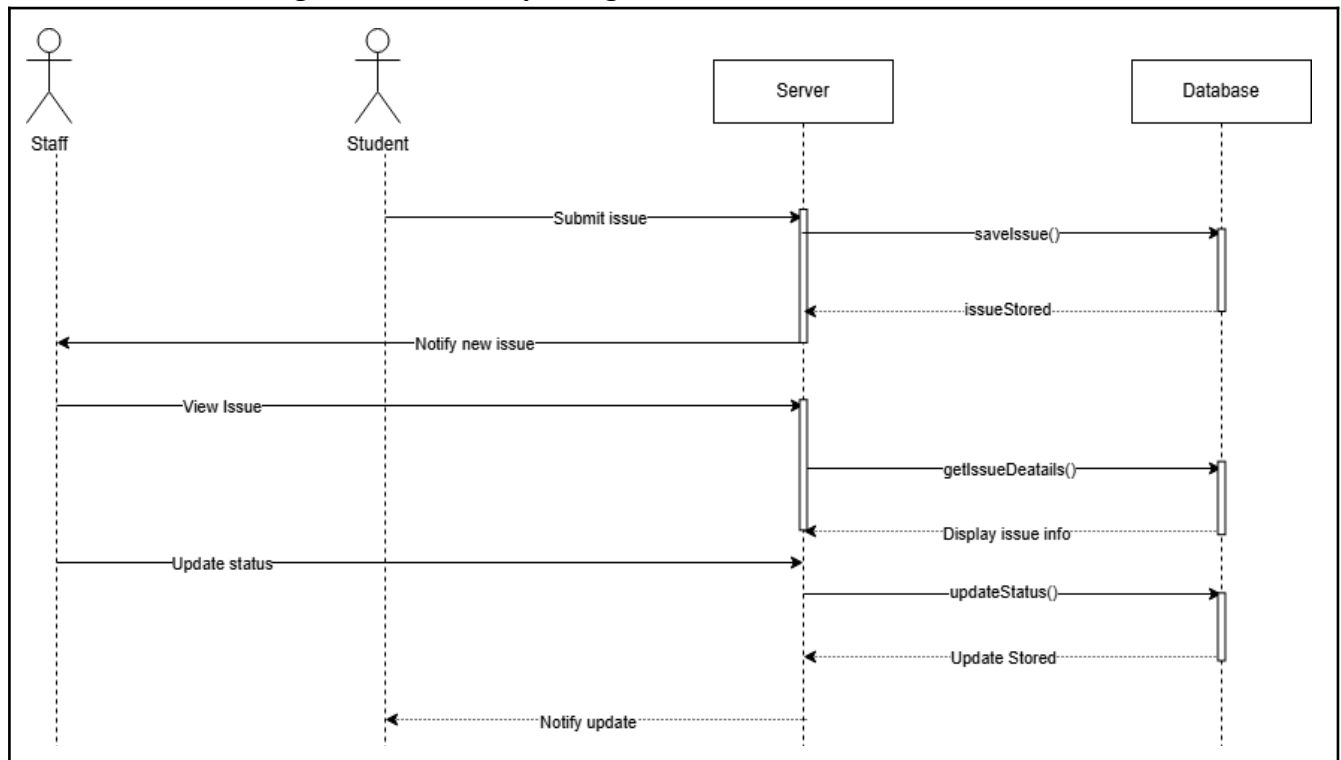


Figure 10.3: Interaction Diagram for Issue Reporting

10.3.2 Interaction Diagram for Course Registration

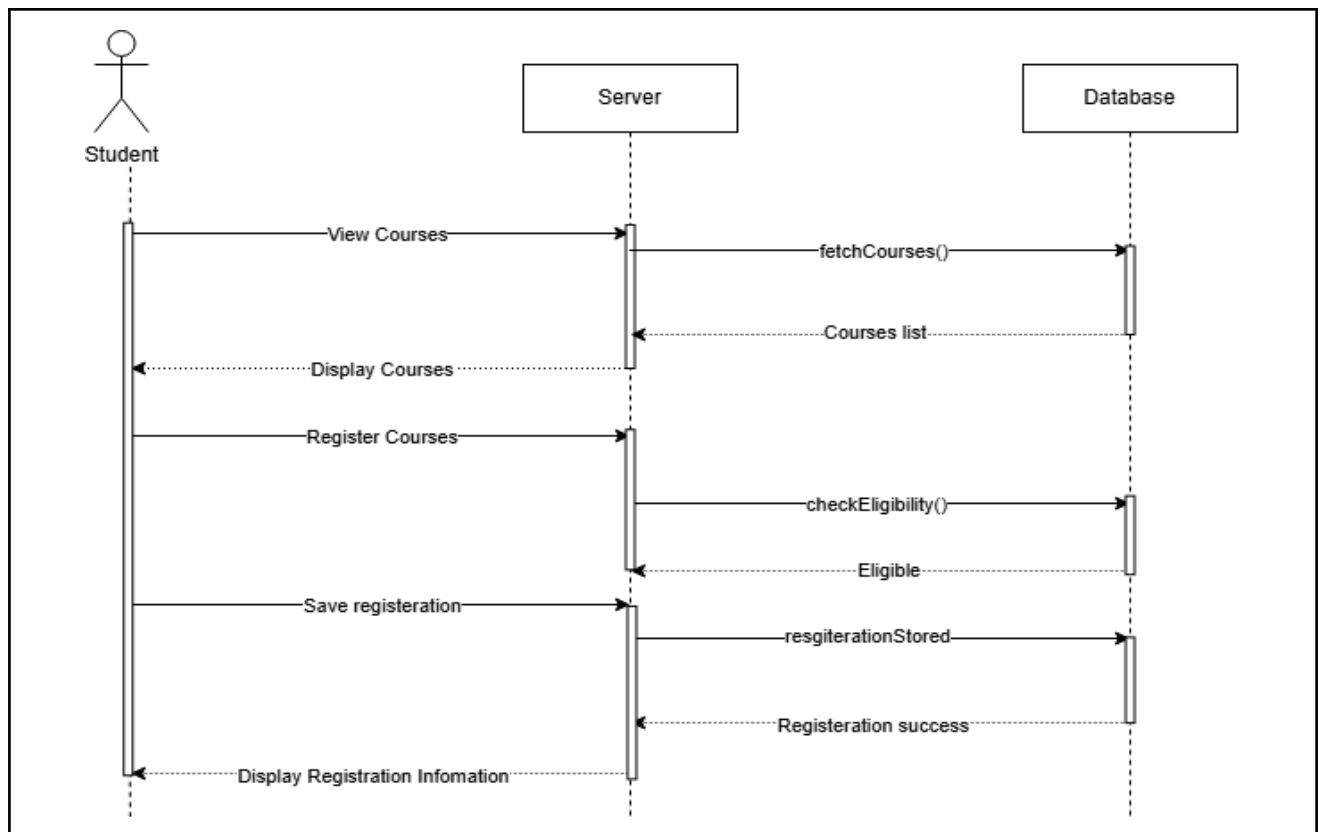


Figure 10.4: Interaction Diagram for Course Registration

10.3.3 Interaction Diagram for Facilities Booking

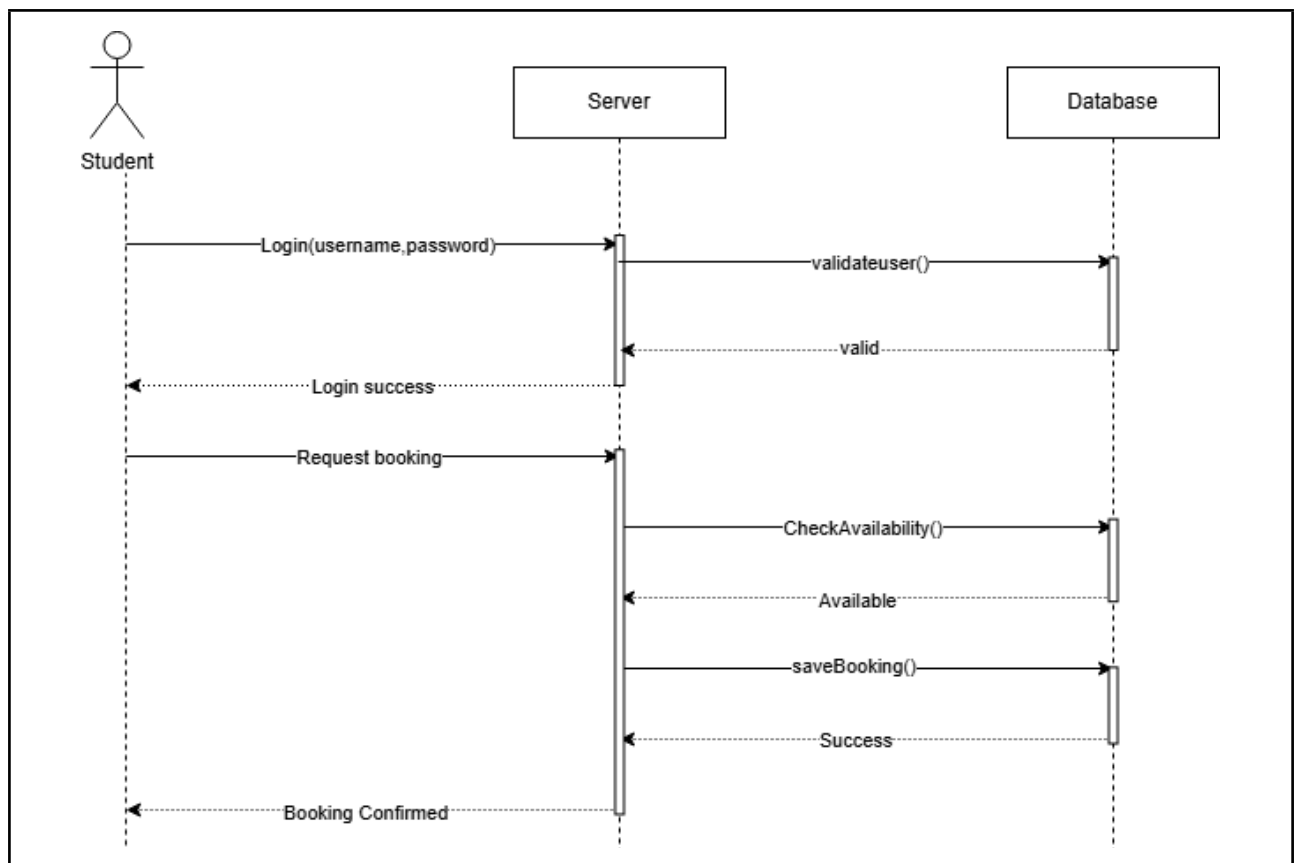


Figure 10.5: Interaction Diagram for Facilities Booking

10.3.4 Interaction Diagram for Admin Staff Add Facility

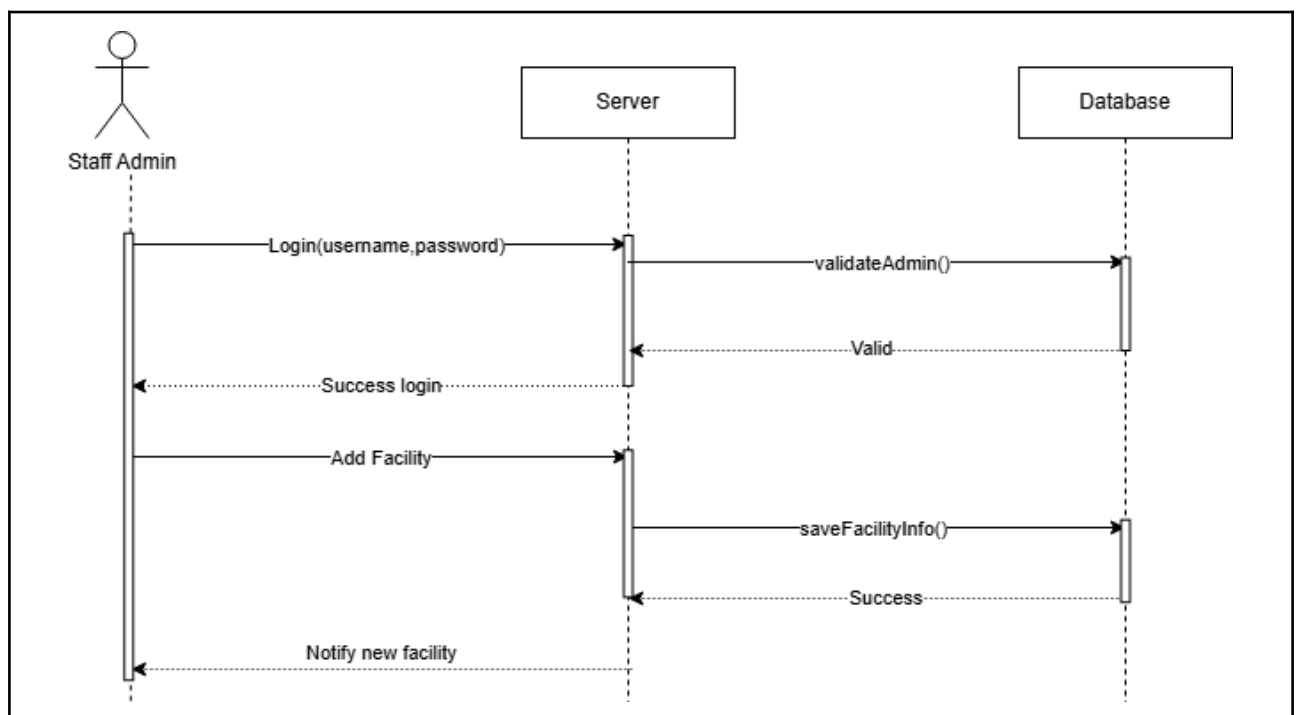


Figure 10.6: Interaction Diagram for Admin Staff Add Facility

10.4 Deployment Diagram

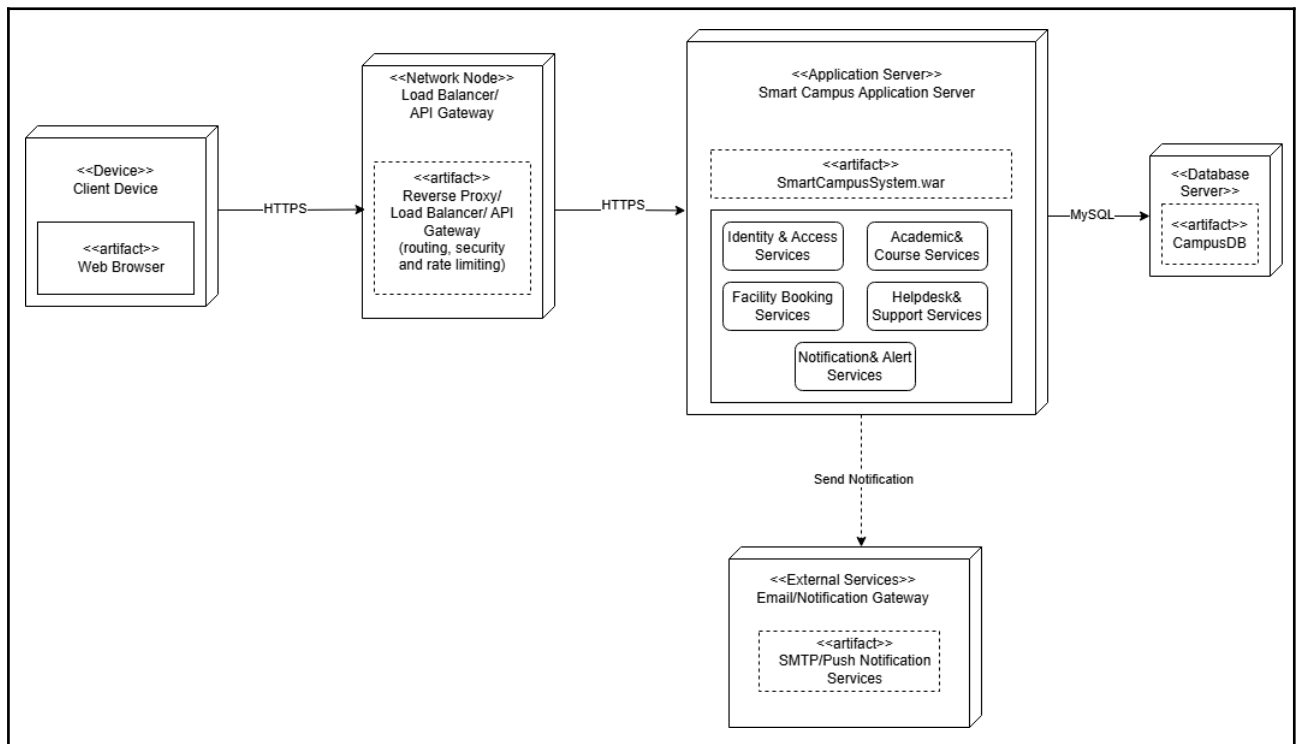


Figure 10.7: Deployment Diagram for Service-Oriented Smart Campus Support System

The diagram shows the architecture of the *Smart Campus Support System*, which follows a service-oriented approach. Users access the system through a web browser, and requests are sent via HTTPS to the API Gateway. This component acts as a load balancer and reverse proxy, handling routing, security, and rate limiting.

The requests are then processed by the *Smart Campus Application Server*, which contains several service modules such as Identity & Access, Academic & Course, Facility Booking, Helpdesk Support, and Notification Services. The server communicates with a MySQL database to store and manage data.

Additionally, the Notification Service connects to an external Email Notification Gateway to send alerts to users. This architecture ensures a secure, scalable, and well-organized system.